Document Title

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1 Introduction

1.1 Matérn covariance function and its parametrizations

From [3], the Matérn covariance function is defined as

$$r(h) = \frac{\sigma^2}{2^{\nu-1}\Gamma(\nu)} \left(\frac{\sqrt{2\nu}h}{\rho} \right)^{\nu} K_\nu \left(\frac{\sqrt{2\nu}h}{\rho} \right) \tag{1}$$

and its corresponding spectral density is

$$\hat{f}(\omega) = \frac{\sigma^2 \sqrt{4\pi}\Gamma(\nu + 1/2)}{\Gamma(\nu)} \left(\frac{2\nu}{\rho^2} \right)^{\nu} \left(\frac{2\nu}{\rho^2} + 4\pi^2\omega^2 \right)^{-(\nu+1/2)} \tag{2}$$

computed by using the Fourier transform pair parametrization

$$\hat{f}(\omega) = \int_{\mathbb{R}} e^{-i2\pi\omega h} r(h) dh \iff r(h) = \int_{\mathbb{R}} e^{i2\pi\omega h} \hat{f}(\omega) d\omega \tag{3}$$

But actually we want the pair parametrization

$$\tilde{f}(\omega) = \int_{\mathbb{R}} e^{-i\omega h} r(h) dh \iff r(h) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{i\omega h} \tilde{f}(\omega) d\omega \tag{4}$$

For that, we simply do

$$\tilde{f}(\omega) = \hat{f}\left(\frac{\omega}{2\pi} \right) = \frac{\sigma^2 \sqrt{4\pi}\Gamma(\nu + 1/2)}{\Gamma(\nu)} \left(\frac{2\nu}{\rho^2} \right)^{\nu} \left(\frac{2\nu}{\rho^2} + \omega^2 \right)^{-(\nu+1/2)} \tag{5}$$

But actually we apparently want this Fourier pair parametrization

$$\overline{f}(\omega) = \mathfrak{F}[r(h)](\omega) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-i\omega h} r(h) dh \iff r(h) = \mathfrak{F}^{-1}[\overline{f}(\omega)](h) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{i\omega h} \overline{f}(\omega) d\omega \tag{6}$$

For that, we simply do

$$\overline{f}(\omega) = \frac{1}{\sqrt{2\pi}} \tilde{f}(\omega) = \frac{\sigma^2 \sqrt{2}\Gamma(\nu + 1/2)}{\Gamma(\nu)} \left(\frac{2\nu}{\rho^2} \right)^{\nu} \left(\frac{2\nu}{\rho^2} + \omega^2 \right)^{-(\nu+1/2)} \tag{7}$$

Recall the scaling property of the Fourier transform given by

$$r(ah) \iff \frac{1}{|a|} \overline{f}\left(\frac{\omega}{a} \right) \tag{8}$$

Now let $\varrho(h) = r(2h)$ and $\kappa = \sqrt{8\nu}/\rho$, then

$$\varrho(h) = \frac{\sigma^2}{2^{\nu-1}\Gamma(\nu)} \left(\frac{\sqrt{8\nu}h}{\rho} \right)^{\nu} K_\nu \left(\frac{\sqrt{8\nu}h}{\rho} \right) \tag{9}$$

$$= \frac{\sigma^2}{2^{\nu-1}\Gamma(\nu)} (\kappa h)^{\nu} K_\nu(\kappa h) \tag{10}$$

which is the familiar form of the Matérn covariance function. Just for the record, if we have it in terms of τ, then

$$\varrho(h) = \frac{1}{2^{\nu-1}\Gamma(\nu)} \frac{\Gamma(\nu)}{\Gamma(\nu + 1/2)\sqrt{4\pi}\kappa^{2\nu}\tau^2} (\kappa h)^{\nu} K_\nu(\kappa h) \tag{11}$$

$$= \frac{1}{2^{\nu-1}\Gamma(\nu + 1/2)\sqrt{4\pi}\kappa^{2\nu}\tau^2} (\kappa h)^{\nu} K_\nu(\kappa h) \tag{12}$$

Setting $\nu = \frac{1}{2}$, we obtain

$$\varrho(h) = \frac{1}{2^{-1/2}\Gamma(1)\sqrt{4\pi}\kappa^{1}\tau^2} (\kappa h)^{1/2} K_{1/2}(\kappa h). \tag{13}$$

Using the identities

$$\Gamma(1) = 1, \tag{14}$$

$$K_{1/2}(x) = \sqrt{\frac{\pi}{2x}} e^{-x}, \tag{15}$$

we get

$$(\kappa h)^{1/2} K_{1/2}(\kappa h) = (\kappa h)^{1/2} \sqrt{\frac{\pi}{2\kappa h}} e^{-\kappa h} = \sqrt{\frac{\pi}{2}}\, e^{-\kappa h}. \tag{16}$$

Substituting back,

$$\varrho(h) = \frac{1}{2^{-1/2}\sqrt{4\pi}\kappa\tau^2} \sqrt{\frac{\pi}{2}}\, e^{-\kappa h}. \tag{17}$$

Now simplify the constants:

$$2^{-1/2}\sqrt{4\pi} = \frac{1}{\sqrt{2}} \cdot 2\sqrt{\pi} = \sqrt{2\pi}, \tag{18}$$

so that

$$\varrho(h) = \frac{1}{\sqrt{2\pi}\,\kappa\tau^2} \sqrt{\frac{\pi}{2}}\, e^{-\kappa h} = \frac{1}{2\kappa\tau^2} e^{-\kappa h}. \tag{19}$$

By the scaling property, the corresponding spectral density to $\varrho(h)$ is $f(\omega) = \frac{1}{2}\overline{f}\left(\frac{\omega}{2}\right)$ given by

$$f(\omega) = \frac{\sigma^2\sqrt{2}\Gamma(\nu + 1/2)}{2\Gamma(\nu)} \left(\frac{2\nu}{\rho^2} \right)^{\nu} \left(\frac{2\nu}{\rho^2} + \frac{\omega^2}{4} \right)^{-(\nu+1/2)} \tag{20}$$

$$= \frac{\sigma^2\sqrt{2}\Gamma(\nu + 1/2)}{2\Gamma(\nu)} \left(\frac{2\nu}{\rho^2} \right)^{\nu} \left(\frac{1}{4} \right)^{-(\nu+1/2)} \left(\frac{8\nu}{\rho^2} + \omega^2 \right)^{-(\nu+1/2)} \tag{21}$$

Since $\left(\frac{1}{4}\right)^{-(\nu+1/2)} = 2 \cdot 4^{\nu}$ and $\kappa^2 = 8\nu/\rho^2$ we have that

$$f(\omega) = \frac{\sigma^2\sqrt{2}\Gamma(\nu+1/2)}{\Gamma(\nu)}\left(\frac{8\nu}{\rho^2}\right)^{\nu}\left(\frac{8\nu}{\rho^2}+\omega^2\right)^{-(\nu+1/2)} \tag{22}$$

$$= \frac{\sigma^2\sqrt{2}\Gamma(\nu+1/2)\kappa^{2\nu}}{\Gamma(\nu)}\left(\kappa^2+\omega^2\right)^{-(\nu+1/2)} \tag{23}$$

Let $A = A_{(\nu,\kappa)} = \dfrac{\sqrt{2}\Gamma(\nu+1/2)\kappa^{2\nu}}{\Gamma(\nu)}$ so this matches with [1]. If $\sigma^2 = \dfrac{\Gamma(\nu)}{\Gamma(\nu+1/2)\sqrt{4\pi}\kappa^{2\nu}\tau^2}$, then $A\sigma^2 = (2\pi)^{-1/2}\tau^{-2}$. The moral of the story is that

$$f(\omega) = A\sigma^2\left(\kappa^2+\omega^2\right)^{-(\nu+1/2)} \iff \varrho(h) = \frac{\sigma^2}{2^{\nu-1}\Gamma(\nu)}(\kappa h)^{\nu}K_{\nu}(\kappa h), \tag{24}$$

Putting in terms of α, we instead have

$$f(\omega) = A\sigma^2\left(\kappa^2+\omega^2\right)^{-\alpha} \iff \varrho(h) = \frac{\sigma^2}{2^{\alpha-1/2-1}\Gamma(\alpha-1/2)}(\kappa h)^{\alpha-1/2}K_{\alpha-1/2}(\kappa h), \tag{25}$$

If we drop σ^2, we still have the Fourier pair

$$f_\star(\omega) = A\left(\kappa^2+\omega^2\right)^{-(\nu+1/2)} \iff \varrho_\star(h) = \frac{1}{2^{\nu-1}\Gamma(\nu)}(\kappa h)^{\nu}K_{\nu}(\kappa h), \tag{26}$$

1.2 Computational quirks

The `MetricGraph` package actually codifies

$$\overline{\varrho}(h;\kappa,\nu,\texttt{cte}) = \frac{\texttt{cte}^2}{2^{\nu-1}\Gamma(\nu)}(\kappa|h|)^\nu K_\nu(\kappa|h|) \tag{27}$$

So if you want something like

$$\frac{c_\alpha\sqrt{\pi}}{\sqrt{1-p_i}}\varrho(h;\kappa,\nu,\sigma) = \frac{c_\alpha\sqrt{\pi}}{\sqrt{1-p_i}}\frac{\sigma^2}{2^{\nu-1}\Gamma(\nu)}(\kappa|h|)^\nu K_\nu(\kappa|h|) \tag{28}$$

computationally you should actually do

$$\sigma^2\overline{\varrho}\left(h;\kappa,\nu,\sqrt{\frac{c_\alpha\sqrt{\pi}}{\sqrt{1-p_i}}}\right) \tag{29}$$

1.3 Preliminaries

We consider the Matérn covariance function

$$\varrho(h;\kappa,\nu,\sigma) = \frac{\sigma^2}{2^{\nu-1}\Gamma(\nu)}(\kappa|h|)^\nu K_\nu(\kappa|h|), \quad \sigma^2 = A^{-1}(2\pi)^{-1/2}\tau^{-2}, \tag{30}$$

where $\rho = \sqrt{8\nu}/\kappa$ and $\nu = \alpha - 1/2$ and the SPDE

$$(\kappa^2 - \Delta_\Gamma)^{\alpha/2}(\tau u) = \mathcal{W}, \qquad \text{on } \Gamma, \tag{31}$$

1.4 Computing spectral density from SPDE

We take the Fourier transform (6) to (31) to get

$$(\kappa^2+\omega^2)^{\alpha/2}\tau\hat{u}(\omega) = \widehat{\mathcal{W}}(\omega) \implies f(\omega) = \frac{1}{\tau^2\sqrt{2\pi}}(\kappa^2+\omega^2)^{-\alpha} \tag{32}$$

Let $L = \kappa^2 - \Delta_\Gamma$ and $B = \kappa^{-2}L$, with $\kappa, \tau > 0$ and $\alpha > 1/2$. The solution u is a centered Gaussian random field with covariance operator $\mathcal{C} = \kappa^{-2\alpha}\tau^{-2}B^{-\alpha}$. For integer $\alpha \geq 1$, the fractional factor is exactly one: set $k = 1$ and omit all residue components, giving the exact covariance $\mathcal{C} = \tau^{-2}L^{-\alpha}$. For noninteger α, we consider the approximation

$$\mathcal{C}_\alpha = \kappa^{-2\alpha}\tau^{-2}B^{-\lfloor\alpha\rfloor}P_m(B)Q_m(B)^{-1} \tag{33}$$

where

$$x^{\{\alpha\}} \approx \frac{p_m(x)}{q_m(x)} = \frac{\sum_{i=0}^m a_i x^i}{\sum_{j=0}^m b_j x^j} = \frac{P_m(x^{-1})}{Q_m(x^{-1})} = r_m(x) \tag{34}$$

or

$$x^{-\{\alpha\}} \approx \frac{p_m(x^{-1})}{q_m(x^{-1})} = \frac{\sum_{i=0}^m a_i x^{m-i}}{\sum_{j=0}^m b_j x^{m-j}} = \frac{P_m(x)}{Q_m(x)} = R_m(x) \tag{35}$$

We further have that

$$\frac{P_m(x)}{Q_m(x)} = \frac{p_m(x^{-1})}{q_m(x^{-1})} = k + \sum_{i=1}^m r_i(x-p_i)^{-1}. \tag{36}$$

Therefore

$$\mathcal{C}_\alpha = \kappa^{-2\alpha}\tau^{-2}B^{-\lfloor\alpha\rfloor}\left(kI + \sum_{i=1}^m r_i(B-p_iI)^{-1}\right) = \kappa^{-2\alpha}\tau^{-2}kB^{-\lfloor\alpha\rfloor} + \sum_{i=1}^m \kappa^{-2\alpha}\tau^{-2}r_iB^{-\lfloor\alpha\rfloor}(B-p_iI)^{-1} \tag{37}$$

Since $k, r_i > 0$ and $p_i < 0$,

$$\mathcal{C}_0 = \kappa^{-2\alpha}\tau^{-2}kB^{-\lfloor\alpha\rfloor} = \kappa^{-2(\alpha-\lfloor\alpha\rfloor)}\tau^{-2}kL^{-\lfloor\alpha\rfloor}$$

and

$$\mathcal{C}_i = \kappa^{-2\alpha}\tau^{-2}r_iB^{-\lfloor\alpha\rfloor}(B-p_iI)^{-1} = \kappa^{-2(\alpha-\lfloor\alpha\rfloor-1)}\tau^{-2}r_iL^{-\lfloor\alpha\rfloor}(L-\kappa^2p_iI)^{-1}$$

are valid covariance operators so that $\mathcal{C}_\alpha = \mathcal{C}_0 + \sum_{i=1}^m \mathcal{C}_i$. Then the approximation u_α has decomposition

$$u_\alpha = u_0 + \sum_{i=1}^m u_i \tag{38}$$

where $u_0 \sim N(0, \mathcal{C}_0)$ and $u_i \sim N(0, \mathcal{C}_i)$, $i = 1, \ldots, m$, are mutually independent. They satisfy the following SPDEs driven by mutually independent Gaussian white noises $\mathcal{W}_0, \ldots, \mathcal{W}_m$:

$$\mathcal{C}_0^{-1/2}u_0 = B^{\lfloor\alpha\rfloor/2}(\tau k^{-1/2}\kappa^\alpha u_0) = L^{\lfloor\alpha\rfloor/2}(\tau k^{-1/2}\kappa^{\alpha-\lfloor\alpha\rfloor}u_0) = \mathcal{W}_0, \qquad \text{on } \Gamma, \tag{39}$$

and

$$\mathcal{C}_i^{-1/2}u_i = (B^{\lfloor\alpha\rfloor}(B-p_iI))^{1/2}(\tau r_i^{-1/2}\kappa^\alpha u_i) = (L^{\lfloor\alpha\rfloor}(L-\kappa^2p_iI))^{1/2}(\tau r_i^{-1/2}\kappa^{\alpha-\lfloor\alpha\rfloor-1}u_i) = \mathcal{W}_i, \qquad \text{on } \Gamma \tag{40}$$

Definition 1.1. u_i in (40) is called a shifted Whittle-Matérn process on Γ.

Remark 1. Observe that (39) are Whittle-Matérn process where the power $\lfloor\alpha\rfloor$ is integer.

Observe now that (30) has spectral density

$$f(\omega) = A\sigma^2(\kappa^2+\omega^2)^{-(\nu+1/2)}, \quad A = \sigma^{-2}(2\pi)^{-1/2}\tau^{-2} \tag{41}$$

which can be also written as

$$f(\omega) = A\sigma^2\kappa^{-2\alpha}(1+\kappa^{-2}\omega^2)^{-\alpha} = A\sigma^2\kappa^{-2\alpha}(1+\kappa^{-2}\omega^2)^{-\lfloor\alpha\rfloor}(1+\kappa^{-2}\omega^2)^{-\{\alpha\}} \tag{42}$$

By (35), we can approximate f by f_α given by

$$f_\alpha(\omega) = A\sigma^2\kappa^{-2\alpha}\frac{1}{(1+\kappa^{-2}\omega^2)^{\lfloor\alpha\rfloor}}\frac{P_m(1+\kappa^{-2}\omega^2)}{Q_m(1+\kappa^{-2}\omega^2)} \tag{43}$$

By (36), we have that

$$f_\alpha(\omega) = A\sigma^2\kappa^{-2\alpha}\frac{1}{(1+\kappa^{-2}\omega^2)^{\lfloor\alpha\rfloor}}\left(k + \sum_{i=1}^m r_i(1+\kappa^{-2}\omega^2-p_i)^{-1}\right) \tag{44}$$

$$= \frac{A\sigma^2\kappa^{-2\alpha}k}{(1+\kappa^{-2}\omega^2)^{\lfloor\alpha\rfloor}} + \sum_{i=1}^m \frac{A\sigma^2\kappa^{-2\alpha}r_i}{(1+\kappa^{-2}\omega^2)^{\lfloor\alpha\rfloor}(1+\kappa^{-2}\omega^2-p_i)} \tag{45}$$

Let

$$f_0(\omega) = \frac{A\sigma^2\kappa^{-2\alpha}k}{(1+\kappa^{-2}\omega^2)^{\lfloor\alpha\rfloor}} \quad \text{and} \quad f_i(\omega) = \frac{A\sigma^2\kappa^{-2\alpha}r_i}{(1+\kappa^{-2}\omega^2)^{\lfloor\alpha\rfloor}(1+\kappa^{-2}\omega^2-p_i)} \tag{46}$$

then we can write

$$f_\alpha(\omega) = f_0(\omega) + \sum_{i=1}^{m} f_i(\omega), \tag{47}$$

which is a valid decomposition of spectral densities. We now can equivalently write

$$f_0(\omega) = \frac{A\sigma^2\kappa^{-2\alpha}k\kappa^{2\lfloor\alpha\rfloor}}{(\kappa^2+\omega^2)^{\lfloor\alpha\rfloor}} \quad \text{and} \quad f_i(\omega) = \frac{A\sigma^2\kappa^{-2\alpha}r_i\kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2+\omega^2)^{\lfloor\alpha\rfloor}(\kappa^2(1-p_i)+\omega^2)} \tag{48}$$

For an integer $q \geq 0$ and $p < 0$, we have the following identity, with sums understood as zero when $q = 0$:

$$\frac{1}{(\kappa^2+\omega^2)^q(\kappa^2(1-p)+\omega^2)} = \frac{1}{(\kappa^2 p)^q(\kappa^2(1-p)+\omega^2)} - \sum_{j=1}^{q} \frac{1}{(\kappa^2 p)^{q+1-j}(\kappa^2+\omega^2)^j} \tag{49}$$

Indeed, let $r = \dfrac{\kappa^2 p}{\kappa^2+\omega^2}$, then $1 - r = \dfrac{\kappa^2(1-p)+\omega^2}{\kappa^2+\omega^2}$. Hence

$$\begin{aligned}
\sum_{j=1}^{q} \frac{1}{(\kappa^2 p)^{q+1-j}(\kappa^2+\omega^2)^j} &= \frac{1}{(\kappa^2 p)^{q+1}} \sum_{j=1}^{q} \left(\frac{\kappa^2 p}{\kappa^2+\omega^2}\right)^j \\
&= \frac{1}{(\kappa^2 p)^{q+1}} \frac{r(1-r^q)}{1-r} \\
&= \frac{1}{(\kappa^2 p)^{q+1}} \frac{\dfrac{\kappa^2 p}{\kappa^2+\omega^2}\left(1 - \left(\dfrac{\kappa^2 p}{\kappa^2+\omega^2}\right)^q\right)}{\dfrac{\kappa^2(1-p)+\omega^2}{\kappa^2+\omega^2}} \\
&= \frac{1 - \left(\dfrac{\kappa^2 p}{\kappa^2+\omega^2}\right)^q}{(\kappa^2 p)^q(\kappa^2(1-p)+\omega^2)} \\
&= \frac{1}{(\kappa^2 p)^q(\kappa^2(1-p)+\omega^2)} - \frac{1}{(\kappa^2+\omega^2)^q(\kappa^2(1-p)+\omega^2)}
\end{aligned}$$

Therefore

$$f_i(\omega) = \frac{A\sigma^2\kappa^{-2\alpha}r_i\kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2 p_i)^{\lfloor\alpha\rfloor}(\kappa^2(1-p_i)+\omega^2)} \tag{50}$$

$$- \sum_{j=1}^{\lfloor\alpha\rfloor} \frac{A\sigma^2\kappa^{-2\alpha}r_i\kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2 p_i)^{\lfloor\alpha\rfloor+1-j}(\kappa^2+\omega^2)^j} \tag{51}$$

$$= \frac{A_{(\nu,\kappa)}\kappa^{-2\alpha}r_i\kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2 p_i)^{\lfloor\alpha\rfloor}}\sigma^2(\kappa^2(1-p_i)+\omega^2)^{-1} \tag{52}$$

$$- \sum_{j=1}^{\lfloor\alpha\rfloor} \frac{A_{(\nu,\kappa)}\kappa^{-2\alpha}r_i\kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2 p_i)^{\lfloor\alpha\rfloor+1-j}}\sigma^2(\kappa^2+\omega^2)^{-j} \tag{53}$$

$$= \frac{A_{(\nu,\kappa)}\kappa^{-2\alpha}r_i\kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2 p_i)^{\lfloor\alpha\rfloor}A_{(1/2,\kappa\sqrt{1-p_i})}}A_{(1/2,\kappa\sqrt{1-p_i})}\sigma^2(\kappa^2(1-p_i)+\omega^2)^{-(1/2+1/2)} \tag{54}$$

$$- \sum_{j=1}^{\lfloor\alpha\rfloor} \frac{A_{(\nu,\kappa)}\kappa^{-2\alpha}r_i\kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2 p_i)^{\lfloor\alpha\rfloor+1-j}A_{(j-1/2,\kappa)}}A_{(j-1/2,\kappa)}\sigma^2(\kappa^2+\omega^2)^{-(j-1/2+1/2)} \tag{55}$$

$$= f_{i0}(\omega) - \sum_{j=1}^{\lfloor\alpha\rfloor} f_{ij}(\omega) \tag{56}$$

$$\begin{cases} \varrho(h) & = \mathfrak{F}^{-1}[f(\omega)](h) = \dfrac{\sigma^2}{2^{\nu-1}\Gamma(\nu)}(\kappa h)^\nu K_\nu(\kappa h) \\ f(\omega) & = A_{(\nu,\kappa)}\sigma^2\left(\kappa^2+\omega^2\right)^{-(\nu+1/2)} \\ A_{(\nu,\kappa)} & = \dfrac{\sqrt{2}\Gamma(\nu+1/2)\kappa^{2\nu}}{\Gamma(\nu)} = \dfrac{\sqrt{2}\Gamma(\alpha)\kappa^{2\nu}}{\Gamma(\alpha-1/2)} \\ \sigma^2 & = A_{(\nu,\kappa)}^{-1}(2\pi)^{-1/2}\tau^{-2} \\ \rho & = \sqrt{8\nu}/\kappa \\ \nu & = \alpha - 1/2 \end{cases} \tag{57}$$

To express f_0 as a Matérn spectrum, we require $\alpha \geq 1$, so that $\lfloor\alpha\rfloor \geq 1$. The case $1/2 < \alpha < 1$ is treated separately in Subsection 2.3. For $\alpha \geq 1$,

$$f_0(\omega) = A_{(\nu,\kappa)}\kappa^{-2\alpha}k\kappa^{2\lfloor\alpha\rfloor}\sigma^2(\kappa^2+\omega^2)^{-\lfloor\alpha\rfloor} = \frac{A_{(\nu,\kappa)}\kappa^{-2\alpha}k\kappa^{2\lfloor\alpha\rfloor}}{A_{(\lfloor\alpha\rfloor-1/2,\kappa)}}A_{(\lfloor\alpha\rfloor-1/2,\kappa)}\sigma^2(\kappa^2+\omega^2)^{-(\lfloor\alpha\rfloor-1/2+1/2)} \tag{58}$$

By taking the inverse Fourier transform, we arrive at

$$\varrho_0(h) = \frac{A_{(\nu,\kappa)}\kappa^{-2\alpha}k\kappa^{2\lfloor\alpha\rfloor}}{A_{(\lfloor\alpha\rfloor-1/2,\kappa)}}\varrho(h;\kappa,\lfloor\alpha\rfloor-1/2,\sigma) \tag{59}$$

and

$$\varrho_i(h) = \frac{A_{(\nu,\kappa)}\kappa^{-2\alpha}r_i\kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2p_i)^{\lfloor\alpha\rfloor}A_{(1/2,\kappa\sqrt{1-p_i})}}\varrho(h;\kappa\sqrt{1-p_i},1/2,\sigma) - \sum_{j=1}^{\lfloor\alpha\rfloor}\frac{A_{(\nu,\kappa)}\kappa^{-2\alpha}r_i\kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2p_i)^{\lfloor\alpha\rfloor+1-j}A_{(j-1/2,\kappa)}}\varrho(h;\kappa,j-1/2,\sigma) \tag{60}$$

Let us simplify the expressions above. We first simplify ϱ_0.

$$\begin{aligned}
\varrho_0(h) &= k\sigma^2\kappa^{-2\alpha}\kappa^{2\lfloor\alpha\rfloor}\frac{A_{(\nu,\kappa)}}{A_{(\lfloor\alpha\rfloor-1/2,\kappa)}}\varrho(h;\kappa,\lfloor\alpha\rfloor-1/2,1) \\
&= k\sigma^2\kappa^{-2\alpha}\kappa^{2\lfloor\alpha\rfloor}\frac{\dfrac{\sqrt{2}\Gamma(\nu+1/2)\kappa^{2\nu}}{\Gamma(\nu)}}{\dfrac{\sqrt{2}\Gamma(\lfloor\alpha\rfloor-1/2+1/2)\kappa^{2(\lfloor\alpha\rfloor-1/2)}}{\Gamma(\lfloor\alpha\rfloor-1/2)}}\varrho(h;\kappa,\lfloor\alpha\rfloor-1/2,1) \\
&= k\sigma^2\kappa^{-2\alpha}\kappa^{2\lfloor\alpha\rfloor}\kappa^{2\nu}\kappa^{-2(\lfloor\alpha\rfloor-1/2)}\frac{\dfrac{\Gamma(\alpha)}{\Gamma(\alpha-1/2)}}{\dfrac{\Gamma(\lfloor\alpha\rfloor)}{\Gamma(\lfloor\alpha\rfloor-1/2)}}\varrho(h;\kappa,\lfloor\alpha\rfloor-1/2,1) \\
&= k\sigma^2\kappa^{2(-\alpha+\lfloor\alpha\rfloor+\nu-\lfloor\alpha\rfloor+1/2)}\frac{\dfrac{\Gamma(\alpha)}{\Gamma(\alpha-1/2)}}{\dfrac{\Gamma(\lfloor\alpha\rfloor)}{\Gamma(\lfloor\alpha\rfloor-1/2)}}\varrho(h;\kappa,\lfloor\alpha\rfloor-1/2,1) \\
&= k\sigma^2\frac{c_\alpha}{c_{\lfloor\alpha\rfloor}}\varrho(h;\kappa,\lfloor\alpha\rfloor-1/2,1) \\
&= k\sigma^2\varrho\left(h;\kappa,\lfloor\alpha\rfloor-1/2,\sqrt{c_\alpha/c_{\lfloor\alpha\rfloor}}\right)
\end{aligned}$$

where $c_a = \Gamma(a)/\Gamma(a-1/2)$. For simplicity, let

$$\varrho_{i0}(h) = \frac{A_{(\nu,\kappa)}\kappa^{-2\alpha}r_i\kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2p_i)^{\lfloor\alpha\rfloor}A_{(1/2,\kappa\sqrt{1-p_i})}}\varrho(h;\kappa\sqrt{1-p_i},1/2,\sigma) \tag{61}$$

and

$$\varrho_{ij}(h) = \frac{A_{(\nu,\kappa)}\kappa^{-2\alpha} r_i \kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2 p_i)^{\lfloor\alpha\rfloor+1-j} A_{(j-1/2,\kappa)}} \varrho(h;\kappa, j-1/2, \sigma) \tag{62}$$

Let us now simplify ϱ_{i0}.

$$\begin{aligned}
\varrho_{i0}(h) &= r_i\sigma^2 \frac{1}{p_i^{\lfloor\alpha\rfloor}} \kappa^{-2\alpha}\kappa^{2\lfloor\alpha\rfloor+2}\kappa^{-2\lfloor\alpha\rfloor} \frac{A_{(\nu,\kappa)}}{A_{(1/2,\kappa\sqrt{1-p_i})}} \varrho(h;\kappa\sqrt{1-p_i}, 1/2, 1) \\
&= r_i\sigma^2 \frac{1}{p_i^{\lfloor\alpha\rfloor}} \kappa^{-2\alpha}\kappa^{2\lfloor\alpha\rfloor+2}\kappa^{-2\lfloor\alpha\rfloor} \frac{\frac{\sqrt{2}\Gamma(\alpha)\kappa^{2\nu}}{\Gamma(\alpha-1/2)}}{\frac{\sqrt{2}\Gamma(1/2+1/2)(\kappa\sqrt{1-p_i})^{2\cdot 1/2}}{\Gamma(1/2)}} \varrho(h;\kappa\sqrt{1-p_i}, 1/2, 1) \\
&= r_i\sigma^2 \frac{1}{p_i^{\lfloor\alpha\rfloor}} \kappa^{2(-\alpha+\lfloor\alpha\rfloor+1-\lfloor\alpha\rfloor+\nu-1/2)} \frac{\frac{\Gamma(\alpha)}{\Gamma(\alpha-1/2)}}{\frac{\Gamma(1/2+1/2)\sqrt{1-p_i}}{\Gamma(1/2)}} \varrho(h;\kappa\sqrt{1-p_i}, 1/2, 1) \\
&= r_i\sigma^2 \frac{1}{p_i^{\lfloor\alpha\rfloor}} \frac{c_\alpha\sqrt{\pi}}{\sqrt{1-p_i}} \varrho(h;\kappa\sqrt{1-p_i}, 1/2, 1) \\
&= r_i\sigma^2 \frac{1}{p_i^{\lfloor\alpha\rfloor}} \varrho\left(h;\kappa\sqrt{1-p_i}, 1/2, \sqrt{c_\alpha\sqrt{\pi}/\sqrt{1-p_i}}\right)
\end{aligned}$$

since $\Gamma(1) = 1$ and $\Gamma(1/2) = \sqrt{\pi}$. Now we simplify ϱ_{ij}.

$$\begin{aligned}
\varrho_{ij}(h) &= r_i\sigma^2\kappa^{2(-\alpha+\lfloor\alpha\rfloor+1-\lfloor\alpha\rfloor-1+j)} \frac{1}{p_i^{\lfloor\alpha\rfloor+1-j}} \frac{A_{(\nu,\kappa)}}{A_{(j-1/2,\kappa)}} \varrho(h;\kappa, j-1/2, 1) \\
&= r_i\sigma^2\kappa^{2(-\alpha+\lfloor\alpha\rfloor+1-\lfloor\alpha\rfloor-1+j)} \frac{1}{p_i^{\lfloor\alpha\rfloor+1-j}} \frac{\frac{\sqrt{2}\Gamma(\alpha)\kappa^{2\nu}}{\Gamma(\alpha-1/2)}}{\frac{\sqrt{2}\Gamma(j)\kappa^{2(j-1/2)}}{\Gamma(j-1/2)}} \varrho(h;\kappa, j-1/2, 1) \\
&= r_i\sigma^2\kappa^{2(-\alpha+\lfloor\alpha\rfloor+1-\lfloor\alpha\rfloor-1+j+\nu-j+1/2)} \frac{1}{p_i^{\lfloor\alpha\rfloor+1-j}} \frac{c_\alpha}{c_j} \varrho(h;\kappa, j-1/2, 1) \\
&= r_i\sigma^2 \frac{1}{p_i^{\lfloor\alpha\rfloor+1-j}} \varrho\left(h;\kappa, j-1/2, \sqrt{c_\alpha/c_j}\right)
\end{aligned}$$

To sum up, we have arrived at

$$\varrho_\alpha(h) = \varrho_0(h) + \sum_{i=1}^{m} \varrho_i(h) \tag{63}$$

where, for $\alpha \geq 1$,

$$\varrho_0(h) = k\sigma^2 \varrho\left(h;\kappa, \lfloor\alpha\rfloor - 1/2, \sqrt{c_\alpha/c_{\lfloor\alpha\rfloor}}\right) \tag{64}$$

and

$$\varrho_i(h) = r_i\sigma^2 \frac{1}{p_i^{\lfloor\alpha\rfloor}} \varrho\left(h;\kappa\sqrt{1-p_i}, 1/2, \sqrt{c_\alpha\sqrt{\pi}/\sqrt{1-p_i}}\right) - \sum_{j=1}^{\lfloor\alpha\rfloor} r_i\sigma^2 \frac{1}{p_i^{\lfloor\alpha\rfloor+1-j}} \varrho\left(h;\kappa, j-1/2, \sqrt{c_\alpha/c_j}\right) \tag{65}$$

In the code, we have the following functions:
- $\varrho_i(h) = r_i\sigma^2$`matern.p(s,t,kappa,p = p, alpha)`.
- $\frac{d^k}{dh^k}\varrho_i(h) = r_i\sigma^2$`matern.p.deriv(s,t,kappa,p = p, alpha, deriv = k)`.
- $\mathbf{r}_i(t_1,t_2) = r_i\sigma^2$`matern.p.joint(s,t,kappa,p,alpha)`.

2 Comparison to rat int paper

The calculation above is a detailed derivation of the partial-fraction identity and inverse Fourier transforms underlying Proposition 5 of [1], equation (9) on page 7 of `rat_int_paper.pdf`; its proof appears on pages 28–29. The final component formulas agree after translating the notation, but some conventions, qualifications, and inconsistencies must be separated. The comparison below refers to the supplied PDF, including its printed formulas, rather than to a potentially revised version of the paper.

2.1 Notation and agreement of the component formulas

The component covariance formulas coincide with Proposition 5 after identifying our residues r_i with the paper's c_i and translating the Matérn parametrization:

$$\varrho(h;\kappa,\nu,s) = \varrho^{\text{paper}}(h;\nu,\kappa,s^2). \tag{66}$$

Our last argument is a standard deviation, whereas the paper's is a variance; this explains the square roots in our final arguments. For later reference, let $q = \lfloor \alpha \rfloor$, $\kappa_i = \kappa\sqrt{1-p_i}$, and $M_{\nu,\lambda}(h) = \varrho(h;\lambda,\nu,1)$. For $q \geq 1$, the common formulas are

$$\varrho_0(h) = k\sigma^2 \frac{c_\alpha}{c_q} M_{q-1/2,\kappa}(h), \tag{67}$$

$$\varrho_i(h) = r_i \sigma^2 c_\alpha \left[\frac{\sqrt{\pi}}{p_i^q\sqrt{1-p_i}} M_{1/2,\kappa_i}(h) - \sum_{j=1}^{q} \frac{M_{j-1/2,\kappa}(h)}{p_i^{q+1-j} c_j} \right]. \tag{68}$$

2.2 Fourier normalization: an inconsistency in the supplied paper

Equation (4) in the paper is wrong: its factor $1/(2\pi)$ must be $1/\sqrt{2\pi}$ to agree with the printed $A = \sqrt{2}\, c_\alpha \kappa^{2\alpha-1}$ and Proposition 5. The corrected formula is

$$r_{m,\alpha}(h) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{i\omega h} f_{m,\alpha}(\omega)\, d\omega. \tag{69}$$

This is exactly the convention used in (6); after this correction, there is no Fourier normalization difference between the two derivations.

2.3 The range $1/2 < \alpha < 1$ and the white-noise component

For $1/2 < \alpha < 1$, we have $q = \lfloor \alpha \rfloor = 0$. Hence the zeroth term of the spectral decomposition becomes

$$f_0(\omega) = \frac{A\sigma^2\kappa^{-2\alpha}k}{(1+\kappa^{-2}\omega^2)^{\lfloor\alpha\rfloor}} = A\sigma^2\kappa^{-2\alpha}k. \tag{70}$$

Under the symmetric Fourier convention, $\mathfrak{F}^{-1}[1](h) = \sqrt{2\pi}\,\delta_0(h)$, since $\int_{\mathbb{R}} e^{i\omega h}\, d\omega = 2\pi\delta_0(h)$. Therefore,

$$\varrho_0(h) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{i\omega h} A\sigma^2\kappa^{-2\alpha}k\, d\omega \tag{71}$$

$$= \sqrt{2\pi} A\sigma^2\kappa^{-2\alpha}k\, \delta_0(h). \tag{72}$$

Using $A = \sqrt{2}\, c_\alpha \kappa^{2\alpha-1}$ and $A\sigma^2 = (2\pi)^{-1/2}\tau^{-2}$ gives

$$\varrho_0(h) = b\,\delta_0(h), \qquad b = \sqrt{2\pi}A\sigma^2\kappa^{-2\alpha}k = \frac{k\sigma^2 c_\alpha\sqrt{4\pi}}{\kappa} = \frac{k}{\tau^2\kappa^{2\alpha}}. \tag{73}$$

2.4 Mapping to unscaled coefficients

With $L = \kappa^2 - \Delta_\Gamma$, $B = \kappa^{-2}L$, and $q = \lfloor \alpha \rfloor$, the identities $B^{-q} = \kappa^{2q}L^{-q}$ and $(B - p_i I)^{-1} = \kappa^2(L - \kappa^2 p_i I)^{-1}$ give the unscaled form

$$\mathcal{C}_\alpha = \tau^{-2}L^{-q}\left(\widetilde{k}I + \sum_{i=1}^{m}\widetilde{r}_i(L - \widetilde{p}_i I)^{-1}\right), \tag{74}$$

where the dimensionless coefficients map to

$$\widetilde{k} = k\kappa^{2q-2\alpha}, \qquad \widetilde{r}_i = r_i\kappa^{2q+2-2\alpha}, \qquad \widetilde{p}_i = \kappa^2 p_i. \tag{75}$$

3 Implementation

3.1 Case $\alpha \in (1/2, 1)$

$$\varrho_0(h) = k\sigma^2\frac{c_\alpha\sqrt{4\pi}}{\kappa}1_{[h=0]} \implies \mathbf{Q}_0 = \mathbf{I}\cdot\frac{1}{(k\sigma^2 c_\alpha\sqrt{4\pi})/\kappa} \tag{76}$$

$$\varrho_i(h) = r_i\frac{c_\alpha\sqrt{\pi}}{\sqrt{1-p_i}}\varrho(h;\kappa\sqrt{1-p_i},1/2,\sigma) \implies \mathbf{Q}_i = \frac{1}{(r_i c_\alpha\sqrt{\pi})/\sqrt{1-p_i}}\begin{cases}\overline{\nu} = 1/2\\ \overline{\kappa} = \kappa\sqrt{1-p_i}\\ \overline{\tau} = \sqrt{\dfrac{\Gamma(\overline{\nu})}{\Gamma(\overline{\nu}+1/2)\sqrt{4\pi}\overline{\kappa}^{2\overline{\nu}}\sigma^2}}\\ \cdot\texttt{Qalpha1}(\overline{\tau},\overline{\kappa})\end{cases} \tag{77}$$

3.2 Case $\alpha \in (1, 2)$

$$\varrho_0(h) = k\frac{c_\alpha}{c_{\lfloor\alpha\rfloor}}\varrho(h;\kappa,\lfloor\alpha\rfloor - 1/2,\sigma) \implies \mathbf{Q}_0 = \frac{1}{kc_\alpha/c_{\lfloor\alpha\rfloor}}\begin{cases}\overline{\nu} = 1/2\\ \overline{\kappa} = \kappa\\ \overline{\tau} = \sqrt{\dfrac{\Gamma(\overline{\nu})}{\Gamma(\overline{\nu}+1/2)\sqrt{4\pi}\overline{\kappa}^{2\overline{\nu}}\sigma^2}}\\ \cdot\texttt{Qalpha1}(\overline{\tau},\overline{\kappa})\end{cases} \tag{78}$$

$$\varrho_i(h) = \varrho_i(h) \implies \mathbf{Q}_i = \frac{1}{r_i\sigma^2}\left(\texttt{matern.p.precision}(\kappa,p_i,\alpha,\ell_e) - f(\texttt{matern.p.joint}(\kappa,p_i,\alpha))\right) \tag{79}$$

4 Integer smoothness case

Consider the equation

$$L^{\alpha/2}(\tau u) = \mathcal{W}, \quad \text{on } [0,\ell], \tag{80}$$

Assume the equation is posed in $\mathbb{R}$ and let

$$\boldsymbol{u}(t) = [u(t), u^{(1)}(t), \ldots, u^{(\alpha-1)}(t)]^\top, \quad t \in \mathbb{R}. \tag{81}$$

Then $\boldsymbol{u}$ has matrix-valued covariance function [2]

$$\boldsymbol{\varrho} : \mathbb{R} \times \mathbb{R} \mapsto \mathbb{R}^{\alpha \times \alpha}, \quad \boldsymbol{\varrho}(t_1, t_2) = \left[\frac{\partial^{i-1}}{\partial t_2^{i-1}} \frac{\partial^{j-1}}{\partial t_1^{j-1}} \varrho(t_1 - t_2) \right]_{ij \in \{1,2,\ldots,\alpha\}}. \tag{82}$$

Consider the vector-valued process

$$\boldsymbol{u}_\ell(t) = [u_\ell(t), u_\ell^{(1)}(t), \ldots, u_\ell^{(\alpha-1)}(t)]^\top, \quad t \in [0, \ell]. \tag{83}$$

Then the covariance structure of $\boldsymbol{u}_\ell$ is given by $\boldsymbol{\varrho}_\ell = \boldsymbol{\varrho}|_{[0,\ell]\times[0,\ell]}$, that is,

$$\boldsymbol{\varrho}_\ell : [0, \ell] \times [0, \ell] \mapsto \mathbb{R}^{\alpha \times \alpha}, \quad \boldsymbol{\varrho}_\ell(t_1, t_2) = \boldsymbol{\varrho}(t_1, t_2). \tag{84}$$

Before introducing the Whittle-Matérn bridge process, we set the notation. Let $\boldsymbol{x}(t) = [x(t), x^{(1)}(t), \ldots, x^{(\alpha-1)}(t)$ denote a vector-valued process in $\mathbb{R}^\alpha$, such as (81) or (83). We introduce the boundary evaluation operator

$$B_\ell^\alpha x = [\boldsymbol{x}(0)^\top, \boldsymbol{x}(\ell)^\top]^\top,$$

that is,

$$B_\ell^\alpha x = [x(0), x^{(1)}(0), \ldots, x^{(\alpha-1)}(0), x(\ell), x^{(1)}(\ell), \ldots, x^{(\alpha-1)}(\ell)]^\top.$$

Proposition 1. (see [2, Proposition 7]) The precision matrix of $\left[\widetilde{\boldsymbol{u}}_\ell(0)^\top, \widetilde{\boldsymbol{u}}_\ell(\ell)^\top\right]^\top$ is given by

$$\widetilde{\mathbf{Q}}_\ell = \mathbf{Q}_\ell - \frac{1}{2} \begin{bmatrix} \boldsymbol{\varrho}_\ell(0,0)^{-1} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\varrho}_\ell(0,0)^{-1} \end{bmatrix}, \tag{85}$$

where $\mathbf{Q}_\ell$ is the precision matrix of $\left[\boldsymbol{u}_\ell(0)^\top, \boldsymbol{u}_\ell(\ell)^\top\right]^\top$.

Let $\Gamma = (\mathcal{V}, \mathcal{E})$ be a compact metric graph. We denote by $\underline{e}$ and $\overline{e}$ the initial and terminal vertices of an edge $e \in \mathcal{E}$. Recall also that a function $u = \{u_e\}_{e\in\mathcal{E}}$ on Γ can be written in the unified form $u(s) = \sum_{e\in\mathcal{E}} \mathbb{I}(s \in e) u_e(s)$, $s \in \Gamma$. In the same spirit, the associated vector-valued function $\boldsymbol{u}(s) = [u(s), u^{(1)}(s), \ldots, u^{(\alpha-1)}(s)]^\top$, $s \in \Gamma$ can be written compactly as

$$\boldsymbol{u}(s) = \sum_{e\in\mathcal{E}} \mathbb{I}(s \in e) \boldsymbol{u}_e(s), \quad s \in \Gamma, \tag{86}$$

where $\boldsymbol{u}_e(s) = [u_e(s), u_e^{(1)}(s), \ldots, u_e^{(\alpha-1)}(s)]^\top$, $s \in e$. Using the notation in (86), we define the vector collecting the boundary values of $\boldsymbol{u}$ across all edges as

$$\mathbf{U} = [\boldsymbol{u}(\underline{e}_1)^\top, \boldsymbol{u}(\overline{e}_1)^\top, \boldsymbol{u}(\underline{e}_2)^\top, \boldsymbol{u}(\overline{e}_2)^\top, \ldots, \boldsymbol{u}(\underline{e}_{|\mathcal{E}|})^\top, \boldsymbol{u}(\overline{e}_{|\mathcal{E}|})^\top]^\top \in \mathbb{R}^{2\alpha|\mathcal{E}|\times 1}. \tag{87}$$

Since $\boldsymbol{u}(s) = \boldsymbol{u}_e(s)$, $s \in e$, we might as well write

$$\mathbf{U} = [\boldsymbol{u}_1(\underline{e}_1)^\top, \boldsymbol{u}_1(\overline{e}_1)^\top, \boldsymbol{u}_2(\underline{e}_2)^\top, \boldsymbol{u}_2(\overline{e}_2)^\top, \ldots, \boldsymbol{u}_{|\mathcal{E}|}(\underline{e}_{|\mathcal{E}|})^\top, \boldsymbol{u}_{|\mathcal{E}|}(\overline{e}_{|\mathcal{E}|})^\top]^\top \in \mathbb{R}^{2\alpha|\mathcal{E}|\times 1}. \tag{88}$$

It follows that

$$\widetilde{\mathbf{U}} \sim \mathsf{N}(\mathbf{0}, \widetilde{\mathbf{Q}}^{-1}), \quad \widetilde{\mathbf{Q}} = \operatorname{diag}(\{\widetilde{\mathbf{Q}}_e\}_{e\in\mathcal{E}}), \tag{89}$$

where $\widetilde{\mathbf{Q}}_e$ denotes the precision matrix given in (85). In (85), the notation $\widetilde{\mathbf{Q}}_\ell$ emphasizes the dependence on the underlying interval $[0, \ell]$. In the present setting, each edge e is identified with the interval $[\underline{e}, \overline{e}]$, which motivates the notation $\widetilde{\mathbf{Q}}_e$.

References

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